

# Lenguaje SQL III

**Alex Di Genova**

**28/05/2026**

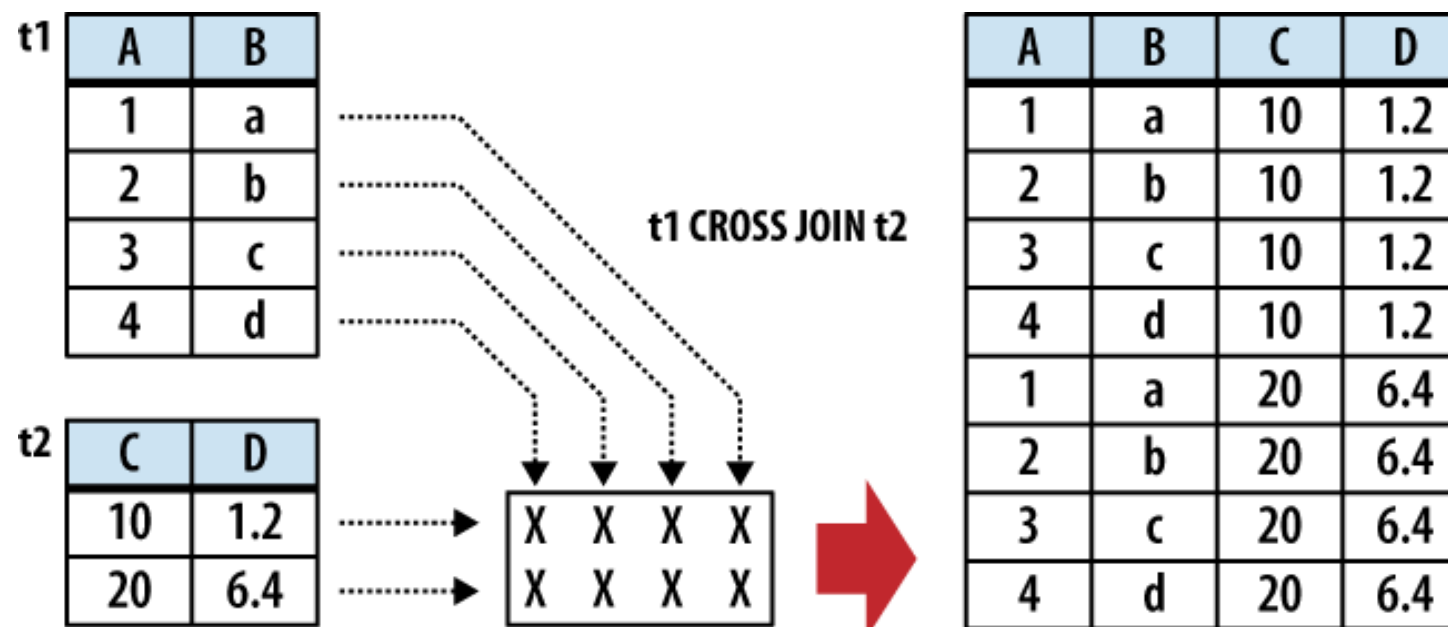
# Contenidos

- JOINS
- subconsultas
- Tecnicas avanzadas

# SQL

## CROSS JOIN

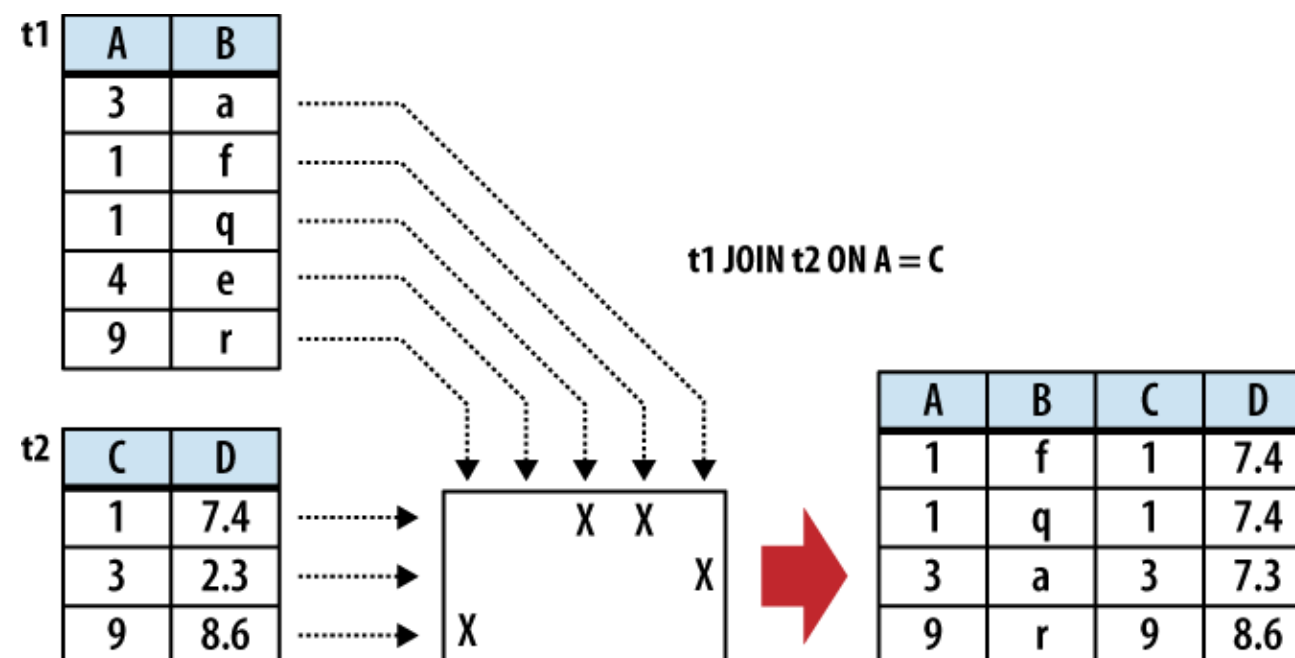
- **SELECT ... FROM t1 CROSS JOIN t2 ...**
- Hace coincidir cada fila de la primera tabla con cada fila de la segunda tabla.
- Si las tablas de entrada tienen x e y columnas, la tabla resultante tendrá x+y columnas.
- Si las tablas de entrada tienen n y m filas, la tabla resultante tendrá n x m filas (producto cartesiano)



# SQL

## INNER JOIN

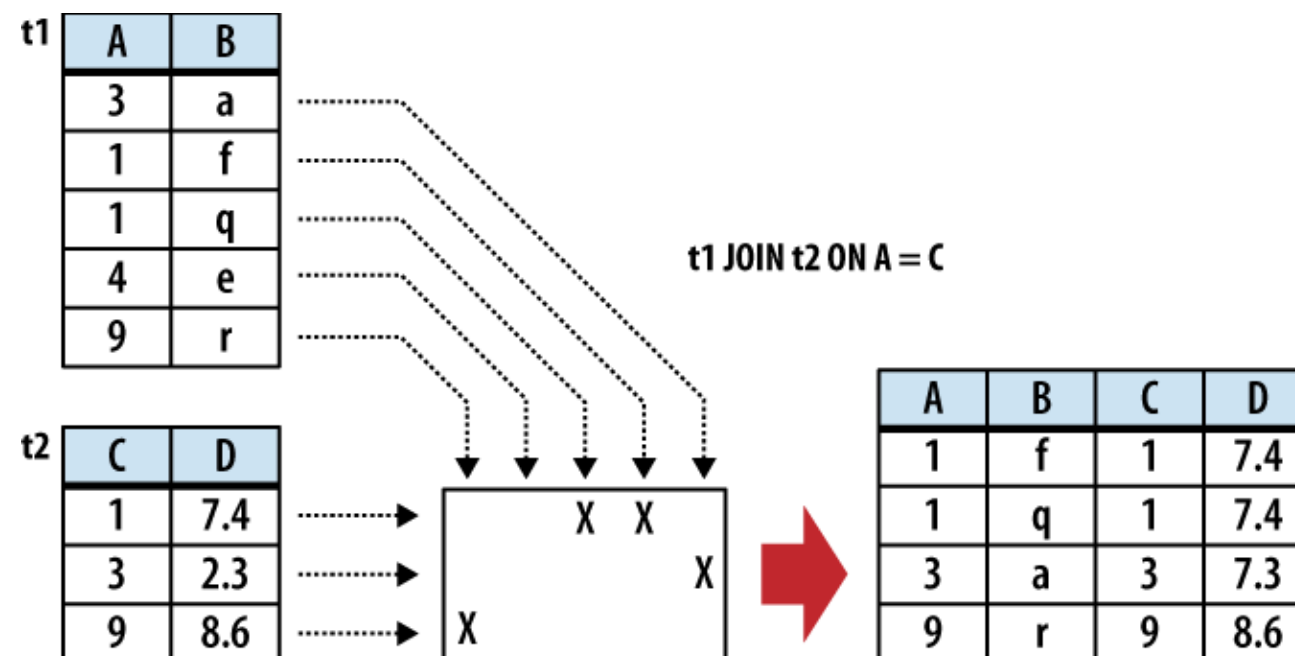
- **SELECT ... FROM t1 JOIN t2 ON expresion\_de\_condicion...**
- La expresion condición es usualmente utilizada para comparar dos columnas de dos tablas diferentes.
- Si las tablas de entrada tienen x e y columnas, la tabla resultante tendrá no más x+y columnas.
- Si las tablas de entrada tienen n y m filas, la tabla resultante tendrá [0,n x m] filas



# SQL

## INNER JOIN

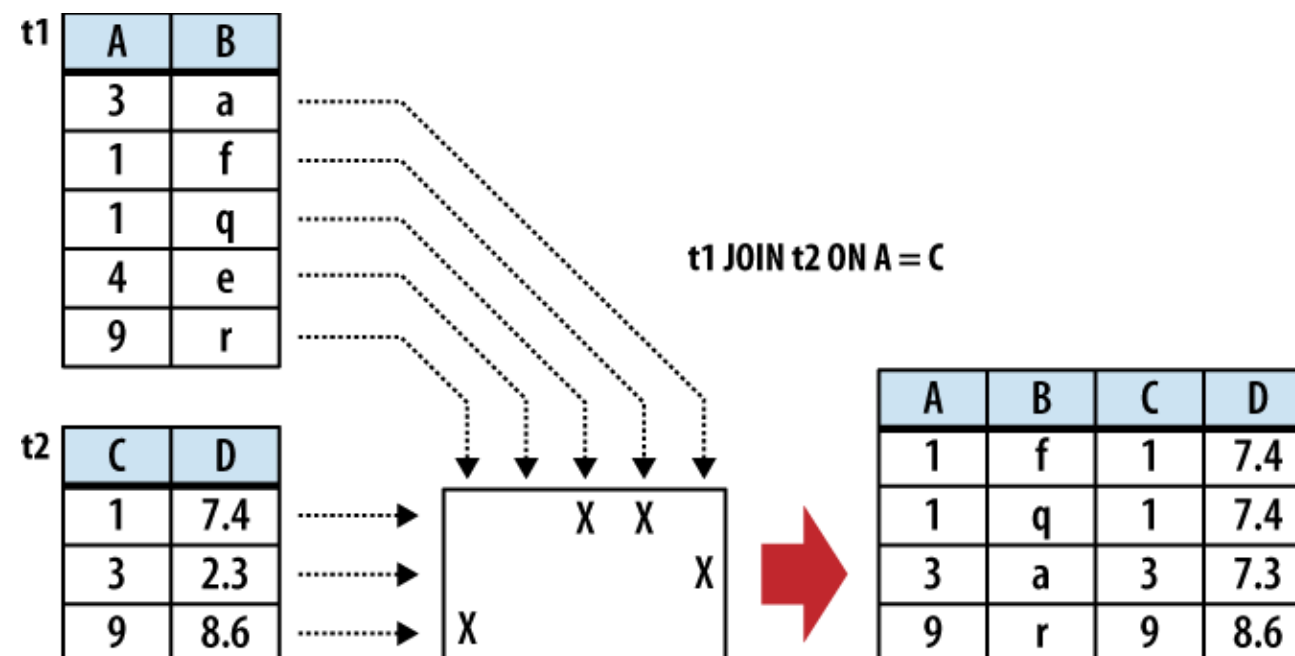
- **SELECT ... FROM t1 INNER JOIN t2 ON expresion\_de\_condicion...**
- **SELECT ... FROM t1 JOIN t2 ON expresion\_de\_condicion...**
- La expresion condición es usualmente utilizada para comparar dos columnas de dos tablas diferentes.
- Si las tablas de entrada tienen x e y columnas, la tabla resultante tendrá no más x+y columnas.
- Si las tablas de entrada tienen n y m filas, la tabla resultante tendrá [0,n x m] filas
- **SELECT ... FROM alumno JOIN curso ON alumno.id=curso.alum\_id...**



# SQL

## INNER JOIN

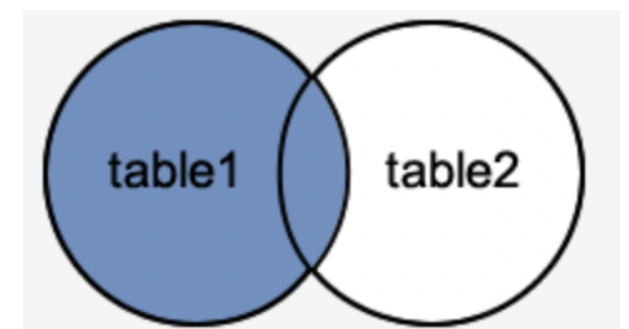
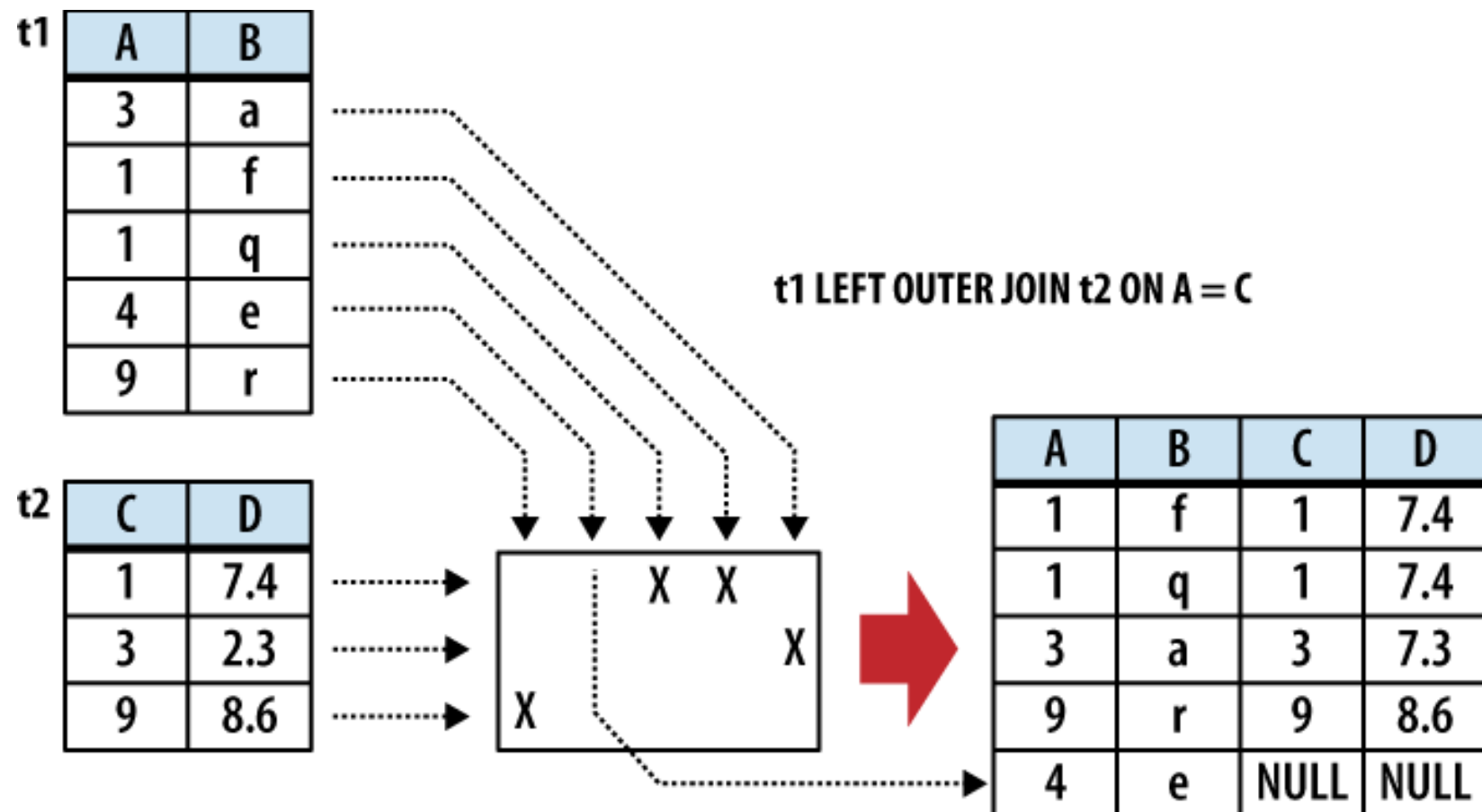
- **SELECT ... FROM** alumno **JOIN** curso **ON** alumno.id=curso.alum\_id...
- **SELECT ... FROM** alumno **JOIN** curso **USING** (aid,...)
  - El atributo debe existir en ambas tablas
- **SELECT ... FROM** alumno **NATURAL JOIN** curso
  - Join automatico en aid



# SQL

## OUTER JOIN

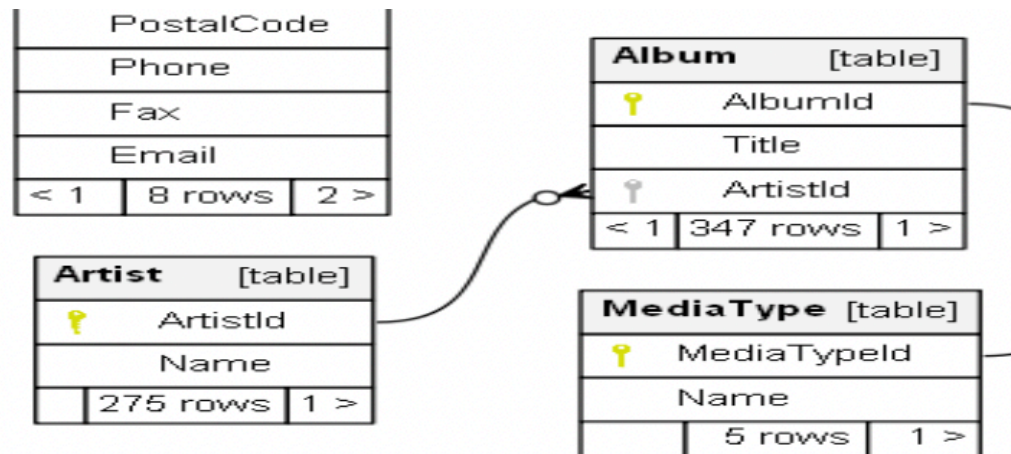
- **SELECT ... FROM** alumno **LEFT OUTER JOIN** curso **ON** alumno.id=curso.alum\_id...
- **Extension del INNER JOIN**
- **OUTER JOINS = LEFT, RIGHT, FULL (SQLite2 solo LEFT)**
- **Condiciones con ON, USING, NATURAL**





# SQL

## JOIN Examples



```
select * from album CROSS JOIN Artist
limit 5;
```

```
select * from album JOIN Artist
ON album.artistId=Artist.artistId limit
5;
```

```
select * from album JOIN Artist USING
(artistId) limit 5;
```

```
select * from album NATURAL JOIN Artist
limit 5;
```

```
1 # CROSS Join example
2 %%sql
3 select * from album CROSS JOIN Artist limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/ Done.

| AlbumId | Title                                 | ArtistId | ArtistId_1 | Name              |
|---------|---------------------------------------|----------|------------|-------------------|
| 1       | For Those About To Rock We Salute You | 1        | 1          | AC/DC             |
| 1       | For Those About To Rock We Salute You | 2        | 2          | Accept            |
| 1       | For Those About To Rock We Salute You | 3        | 3          | Aerosmith         |
| 1       | For Those About To Rock We Salute You | 4        | 4          | Alanis Morissette |
| 1       | For Those About To Rock We Salute You | 5        | 5          | Alice In Chains   |

```
1 #INNER Join
2 %%sql
3 select * from album JOIN Artist ON album.artistId=Artist.artistId
limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/ Done.

| AlbumId | Title                                 | ArtistId | ArtistId_1 | Name      |
|---------|---------------------------------------|----------|------------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | 1          | AC/DC     |
| 2       | Balls to the Wall                     | 2        | 2          | Accept    |
| 3       | Restless and Wild                     | 2        | 2          | Accept    |
| 4       | Let There Be Rock                     | 1        | 1          | AC/DC     |
| 5       | Big Ones                              | 3        | 3          | Aerosmith |

```
1 #SELECT ... FROM alumnno JOIN curso USING (aid,...)
2 %%sql
3 select * from album JOIN Artist USING (artistId) limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/ Done.

| AlbumId | Title                                 | ArtistId | Name      |
|---------|---------------------------------------|----------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC     |
| 2       | Balls to the Wall                     | 2        | Accept    |
| 3       | Restless and Wild                     | 2        | Accept    |
| 4       | Let There Be Rock                     | 1        | AC/DC     |
| 5       | Big Ones                              | 3        | Aerosmith |

```
1 #SELECT ... FROM alumnno NATURAL JOIN curso (artistId)
2 %%sql
3 select * from album NATURAL JOIN Artist limit 5;
```

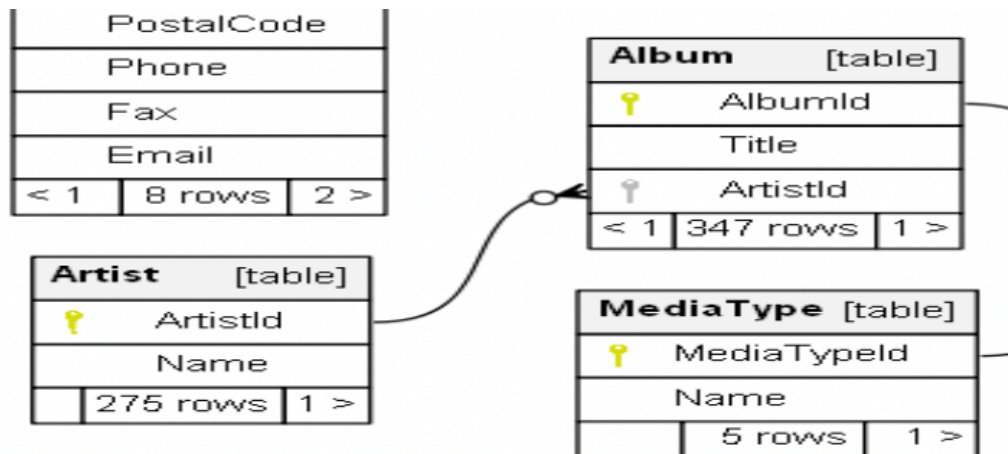
\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/ Done.

| AlbumId | Title                                 | ArtistId | Name      |
|---------|---------------------------------------|----------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC     |
| 2       | Balls to the Wall                     | 2        | Accept    |
| 3       | Restless and Wild                     | 2        | Accept    |
| 4       | Let There Be Rock                     | 1        | AC/DC     |
| 5       | Big Ones                              | 3        | Aerosmith |



# SQL

## JOIN Examples



```
select * from album LEFT OUTER JOIN
Artist limit 5;
```

```
select * from album RIGHT OUTER JOIN
Artist limit 5;
```

```
select * from Artist LEFT OUTER JOIN
album limit 5;
```

```
1 %%sql
2 select * from album LEFT OUTER JOIN Artist USING (artistId) limit 10;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook  
Done.

| AlbumId | Title                                 | ArtistId | Name                 |
|---------|---------------------------------------|----------|----------------------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC                |
| 2       | Balls to the Wall                     | 2        | Accept               |
| 3       | Restless and Wild                     | 2        | Accept               |
| 4       | Let There Be Rock                     | 1        | AC/DC                |
| 5       | Big Ones                              | 3        | Aerosmith            |
| 6       | Jagged Little Pill                    | 4        | Alanis Morissette    |
| 7       | Facelift                              | 5        | Alice In Chains      |
| 8       | Warner 25 Anos                        | 6        | Antônio Carlos Jobim |
| 9       | Plays Metallica By Four Cellos        | 7        | Apocalyptica         |
| 10      | Audioslave                            | 8        | Audioslave           |

```
1 %%sql
2 select * from album RIGHT OUTER JOIN Artist limit 10;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook  
(sqlite3.OperationalError) RIGHT and FULL OUTER JOINS are not currently su  
[SQL: select \* from album RIGHT OUTER JOIN Artist limit 10;]  
(Background on this error at: <https://sqlalche.me/e/14/e3q8>)

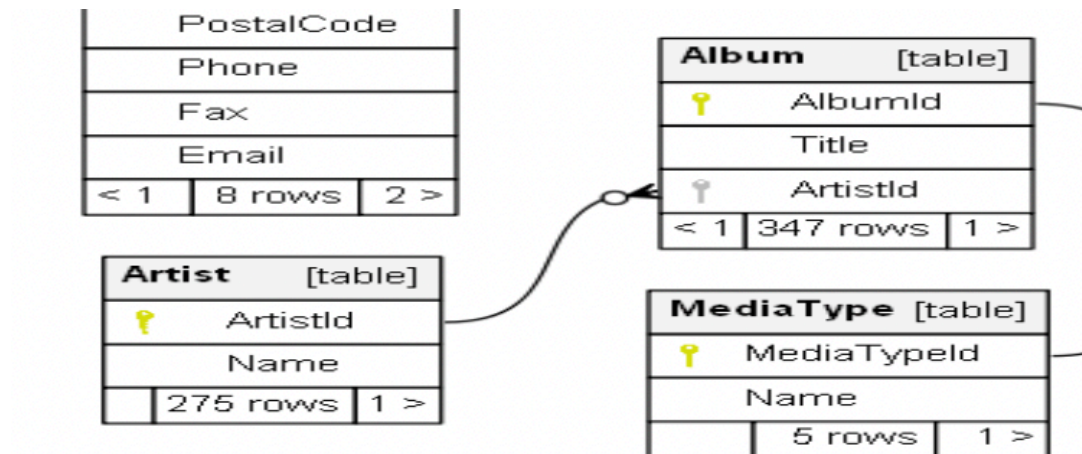
```
1 %%sql
2 select * from Artist LEFT OUTER JOIN album USING (artistId) limit 10;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook  
Done.

| ArtistId | Name                 | AlbumId | Title                                 |
|----------|----------------------|---------|---------------------------------------|
| 1        | AC/DC                | 1       | For Those About To Rock We Salute You |
| 1        | AC/DC                | 4       | Let There Be Rock                     |
| 2        | Accept               | 2       | Balls to the Wall                     |
| 2        | Accept               | 3       | Restless and Wild                     |
| 3        | Aerosmith            | 5       | Big Ones                              |
| 4        | Alanis Morissette    | 6       | Jagged Little Pill                    |
| 5        | Alice In Chains      | 7       | Facelift                              |
| 6        | Antônio Carlos Jobim | 8       | Warner 25 Anos                        |
| 6        | Antônio Carlos Jobim | 34      | Chill: Brazil (Disc 2)                |
| 7        | Apocalyptica         | 9       | Plays Metallica By Four Cellos        |

# SQL

## Notación JOINS



```
select * from album JOIN Artist
ON album.artistId=Artist.artistId limit
5;
```

```
select * from album JOIN Artist USING
(artistId) limit 5;
```

```
select * from album NATURAL JOIN Artist
limit 5;
```

```
select * from album, Artist WHERE
album.artistId=Artist.artistId limit 5;
```

```
3 select * from album JOIN Artist ON album.artistId=Artist.artistId limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_Sqlite.sqlite  
Done.

| AlbumId | Title                                 | ArtistId | ArtistId_1 | Name      |
|---------|---------------------------------------|----------|------------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | 1          | AC/DC     |
| 2       | Balls to the Wall                     | 2        | 2          | Accept    |
| 3       | Restless and Wild                     | 2        | 2          | Accept    |
| 4       | Let There Be Rock                     | 1        | 1          | AC/DC     |
| 5       | Big Ones                              | 3        | 3          | Aerosmith |

```
1 #SELECT ... FROM alumno JOIN curso USING (aid,...)
2 %%sql
3 select * from album JOIN Artist USING (artistId) limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_Sqlite.sqlite  
Done.

| AlbumId | Title                                 | ArtistId | Name      |
|---------|---------------------------------------|----------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC     |
| 2       | Balls to the Wall                     | 2        | Accept    |
| 3       | Restless and Wild                     | 2        | Accept    |
| 4       | Let There Be Rock                     | 1        | AC/DC     |
| 5       | Big Ones                              | 3        | Aerosmith |

```
1 #SELECT ... FROM alumno NATURAL JOIN curso (artistId)
2 %%sql
3 select * from album NATURAL JOIN Artist limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_Sqlite.sqlite  
Done.

| AlbumId | Title                                 | ArtistId | Name      |
|---------|---------------------------------------|----------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC     |
| 2       | Balls to the Wall                     | 2        | Accept    |
| 3       | Restless and Wild                     | 2        | Accept    |
| 4       | Let There Be Rock                     | 1        | AC/DC     |
| 5       | Big Ones                              | 3        | Aerosmith |

```
1 %%sql
2 select * from album, Artist WHERE album.artistId=Artist.artistId limit 5;
```

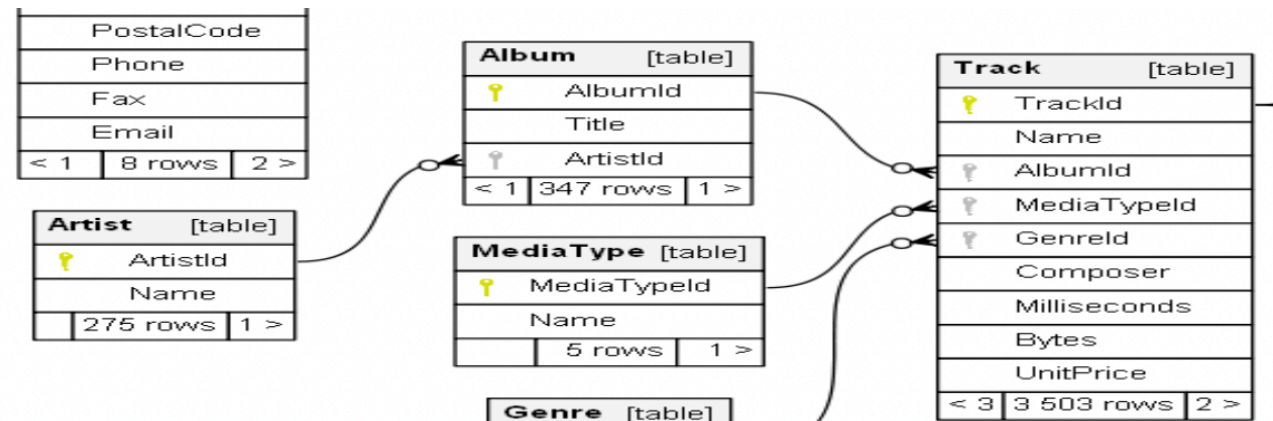
\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_Sqlite.sqlite  
Done.

| AlbumId | Title                                 | ArtistId | ArtistId_1 | Name   |
|---------|---------------------------------------|----------|------------|--------|
| 1       | For Those About To Rock We Salute You | 1        | 1          | AC/DC  |
| 2       | Balls to the Wall                     | 2        | 2          | Accept |



# SQL

## Multiples JOIN (JOIN & JOIN & JOIN...)



```
select * from album JOIN Artist USING (artistId) limit 5;
```

```
select * from album JOIN Artist USING (artistId) JOIN Track USING (albumid) limit 5;
```

```
[12] 1 %%sql
      2 select * from album JOIN Artist USING (artistId) limit 5;
```

```
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook_Sqlite.sqlite
Done.
```

| AlbumId | Title                                 | ArtistId | Name      |
|---------|---------------------------------------|----------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC     |
| 2       | Balls to the Wall                     | 2        | Accept    |
| 3       | Restless and Wild                     | 2        | Accept    |
| 4       | Let There Be Rock                     | 1        | AC/DC     |
| 5       | Big Ones                              | 3        | Aerosmith |

```
1 %%sql
2 select * from album JOIN Artist USING (artistId) JOIN Track USING (albumid) limit 5;
```

```
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook_Sqlite.sqlite
Done.
```

| AlbumId | Title                                 | ArtistId | Name   | TrackId | Name_1                                  | MediaTypeId | GenreId | Composer                                                                | Milliseconds |
|---------|---------------------------------------|----------|--------|---------|-----------------------------------------|-------------|---------|-------------------------------------------------------------------------|--------------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC  | 1       | For Those About To Rock (We Salute You) | 1           | 1       | Angus Young, Malcolm Young, Brian Johnson                               | 343719       |
| 2       | Balls to the Wall                     | 2        | Accept | 2       | Balls to the Wall                       | 2           | 1       | None                                                                    | 342562       |
| 3       | Restless and Wild                     | 2        | Accept | 3       | Fast As a Shark                         | 2           | 1       | F. Baltes, S. Kaufman, U. Dirkschneider & W. Hoffman                    | 230619       |
| 3       | Restless and Wild                     | 2        | Accept | 4       | Restless and Wild                       | 2           | 1       | F. Baltes, R.A. Smith-Diesel, S. Kaufman, U. Dirkschneider & W. Hoffman | 252051       |
| 3       | Restless and Wild                     | 2        | Accept | 5       | Princess of the Dawn                    | 2           | 1       | Deaffy & R.A. Smith-Diesel                                              | 375418       |

# Subconsultas en SQL

## SELECT (SELECT)

- Consultas conteniendo subconsultas
- Subconsultas pueden ser usadas casi en cualquier lugar de las consultas.
- Select .. from where x IN ( select ... )

```
select name , AlbumId
from track
where AlbumId in (
    select AlbumId
    from album
    where title = "Californication")
limit 10;
```

```
1 #Obtener los nombres de todas las canciones del álbum "Californication".
2 %%sql
3 select name , AlbumId
4 from track
5 where AlbumId in (
6     select AlbumId
7     from album
8     where title = "Californication")
9 limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_S  
Done.

| Name              | AlbumId |
|-------------------|---------|
| Around The World  | 195     |
| Parallel Universe | 195     |
| Scar Tissue       | 195     |
| Otherside         | 195     |
| Get On Top        | 195     |

```
1 %%sql
2 select name , AlbumId
3 from track
4 where AlbumId not in (
5     select AlbumId
6     from album
7     where title = "Californication")
8 limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_S  
Done.

| Name                                    | AlbumId |
|-----------------------------------------|---------|
| For Those About To Rock (We Salute You) | 1       |
| Balls to the Wall                       | 2       |
| Fast As a Shark                         | 3       |
| Restless and Wild                       | 3       |
| Princess of the Dawn                    | 3       |

# SQL

## Tecnicas avanzadas

- Subconsultas (IN)
  - SELECT \* from X NATURAL JOIN Y;
  - SELECT \* from X NATURAL JOIN (SELECT \* FROM Y);

```
select * from album NATURAL JOIN  
Artist limit 5;
```

```
select * from album NATURAL JOIN  
(select * from Artist) limit 5;
```

col [not] IN ( test1, test2, test3 )

col [not] IN ( SELECT c FROM t )

col [not] IN temp.in\_test

```
1 %%sql  
2 select * from album NATURAL JOIN Artist limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/ChinookDatabase.sqlite Done.

| AlbumId | Title                                 | ArtistId | Name      |
|---------|---------------------------------------|----------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC     |
| 2       | Balls to the Wall                     | 2        | Accept    |
| 3       | Restless and Wild                     | 2        | Accept    |
| 4       | Let There Be Rock                     | 1        | AC/DC     |
| 5       | Big Ones                              | 3        | Aerosmith |

```
1 %%sql  
2 select * from album NATURAL JOIN (select * from Artist) limit 5;
```

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/ChinookDatabase.sqlite Done.

| AlbumId | Title                                 | ArtistId | Name      |
|---------|---------------------------------------|----------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | AC/DC     |
| 2       | Balls to the Wall                     | 2        | Accept    |
| 3       | Restless and Wild                     | 2        | Accept    |
| 4       | Let There Be Rock                     | 1        | AC/DC     |
| 5       | Big Ones                              | 3        | Aerosmith |

# SQL

## Técnicas avanzadas

Combinación de multiples select

SELECT .. FROM .. WHERE ...  
GROUP BY ... HAVING ...

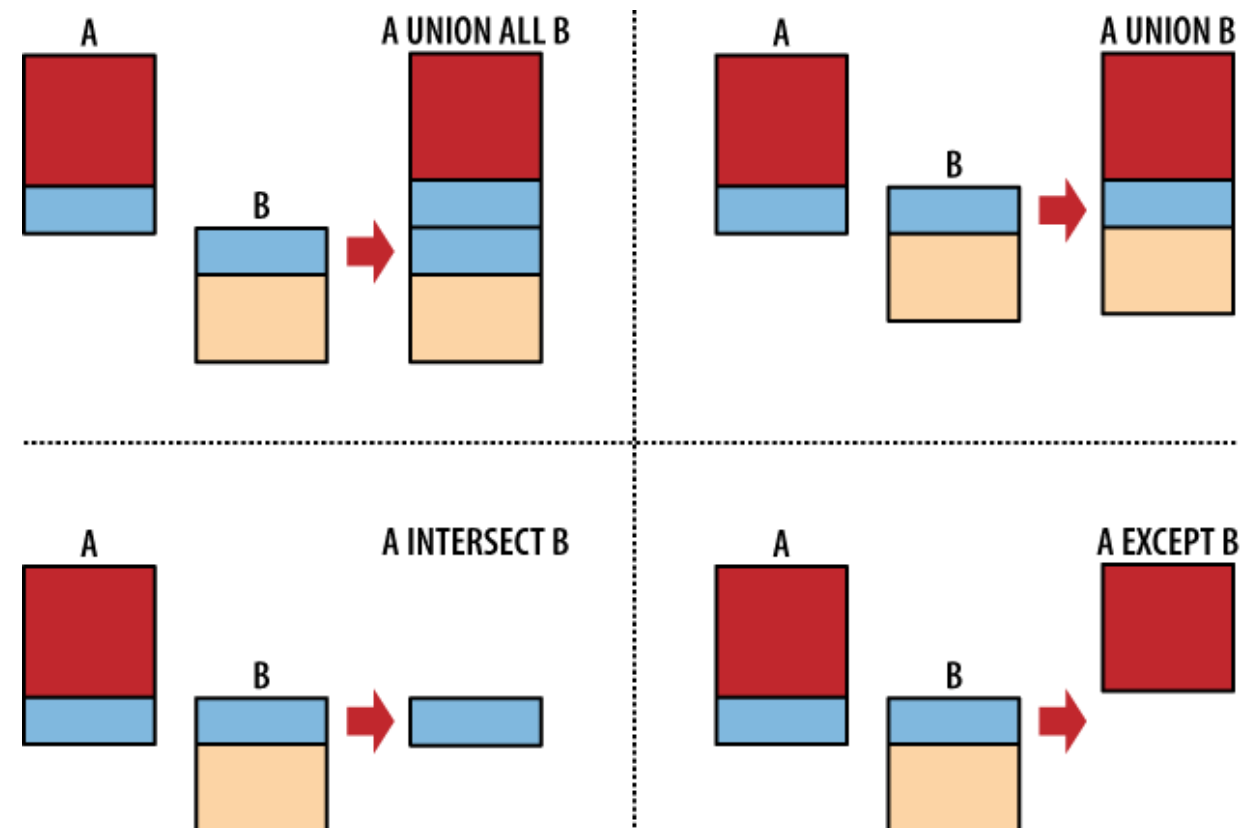
[Operador de mezcla]

SELECT .. FROM .. WHERE ...  
GROUP BY ... HAVING ...

[...]

ORDER BY ... LIMIT ....

UNION ALL, UNION, INTERSECT,  
EXCEPT





# Algebra relacional

## UNION, INTERSECTION, DIFFERENCE, PRODUCT

```
1 %%sql
2 drop table if exists R;
3 drop table if exists S;
4 -- Relación R --
5 CREATE TABLE R (
6 a_id INTEGER,
7 b INTEGER);
8 -- Relación S --
9 CREATE TABLE S (
10 a_id INTEGER,
11 b INTEGER);
12
13 -- valores en R --
14 INSERT INTO R(a_id,b) VALUES
15 (1,10),
16 (2,20),
17 (2,31),
18 (4,32);
19 -- valores en S --
20 INSERT INTO S(a_id,b) VALUES
21 (4,32),
22 (5,50),
23 (7,51);
```

$$R \cup S$$

```
1 %%sql
2 select * from R
3 UNION ALL
4 select * from S;
```

\* sqlite:///test.db  
Done.

| a_id | b  |
|------|----|
| 1    | 10 |
| 2    | 20 |
| 2    | 31 |
| 4    | 32 |
| 4    | 32 |
| 5    | 50 |
| 7    | 51 |

$$R \cap S$$

```
1 %%sql
2 select * from R
3 INTERSECT
4 select * from S;
```

\* sqlite:///test.db  
Done.

| a_id | b  |
|------|----|
| 4    | 32 |

$$R \times S$$

```
1 %%sql
2 select * from R,S;
```

\* sqlite:///test.db  
Done.

| a_id | b  | a_id_1 | b_1 |
|------|----|--------|-----|
| 1    | 10 | 4      | 32  |
| 1    | 10 | 5      | 50  |
| 1    | 10 | 7      | 51  |
| 2    | 20 | 4      | 32  |
| 2    | 20 | 5      | 50  |
| 2    | 20 | 7      | 51  |
| 2    | 31 | 4      | 32  |
| 2    | 31 | 5      | 50  |
| 2    | 31 | 7      | 51  |
| 4    | 32 | 4      | 32  |
| 4    | 32 | 5      | 50  |
| 4    | 32 | 7      | 51  |

$$R - S$$

```
1 %%sql
2 select * from R
3 EXCEPT
4 select * from S;
```

\* sqlite:///test.db  
Done.

| a_id | b  |
|------|----|
| 1    | 10 |
| 2    | 20 |
| 2    | 31 |

# Complex SQL queries

## Que hacen las siguientes consultas?

```
%%sql
```

```
SELECT ar.Name AS Artist, COUNT(DISTINCT(tr.TrackId)) AS TracksSold,  
SUM(ii.UnitPrice) AS Revenue  
FROM artist ar  
INNER JOIN album al ON ar.ArtistId = al.ArtistId  
INNER JOIN track tr ON al.AlbumId = tr.AlbumId  
INNER JOIN invoiceline ii ON tr.TrackId = ii.TrackId  
INNER JOIN invoice i ON ii.InvoiceId = i.InvoiceId  
WHERE strftime('%Y', i.InvoiceDate) = '2013'  
GROUP BY ar.Name  
ORDER BY TracksSold DESC  
LIMIT 5;
```

Identifica los 5 artistas más populares entre los clientes de la tienda de música de Chinook en 2013.

```
%%sql
```

```
SELECT c.FirstName || ' ' || c.LastName AS CustomerName,  
      (SELECT COUNT(*) FROM invoiceline WHERE InvoiceId IN  
        (SELECT InvoiceId FROM invoice WHERE CustomerId = c.CustomerId)) AS TotalTracksPurchased,  
      (SELECT COUNT(DISTINCT AlbumId) FROM  
        (SELECT AlbumId FROM invoiceline INNER JOIN track ON invoiceline.TrackId = track.TrackId  
          WHERE InvoiceId IN (SELECT InvoiceId FROM invoice WHERE CustomerId = c.CustomerId))) AS  
TotalAlbumsPurchased,  
      (SELECT SUM(Total) FROM invoice WHERE CustomerId = c.CustomerId) AS TotalSpent  
FROM customer c;
```



# SQL

## SELECT

- **SELECT** pipeline

**SELECT** [DISTINCT] *select\_header*

**FROM** *tablas*

**WHERE** *expresion\_filtrado*

**GROUP BY** *expresion\_de\_agrupamiento*

**HAVING** *expresion\_filtrado*

**ORDER BY** *expresion\_orden*

**LIMIT** *contador*

*OFFSET indice*

# Practicar en GoogleColab!!!

The screenshot shows a Google Colab notebook interface. The browser tabs include 'The SQLite Query Optimizer', 'SQL\_IV\_chinook\_db.ipynb', 'Home / Twitter', 'join sqlite - Google Search', and 'SQLite: Joins'. The notebook title is 'SQL\_IV\_chinook\_db.ipynb' and it has a star icon. The menu bar includes 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. The left sidebar shows a file explorer with a folder icon and a search icon. The main area displays the following content:

Generated by SchemaSpy

In the above ER diagram, the tiny vertical key icon indicates a column is a primary key. A primary key is a column (or set of columns) whose values uniquely identify every row in a table. For example, `Employeeid` is the primary key in the `Employee` table.

The relationship icon indicates a foreign key constraint and a one-to-many relationship.

▼ SQL con Chinook

▼ Joins

1 # CROSS Join example  
2 %%sql  
3 select \* from album CROSS JOIN Artist limit 5;

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_Sqlite.sqlite  
Done.

| AlbumId | Title                                 | ArtistId | ArtistId_1 | Name              |
|---------|---------------------------------------|----------|------------|-------------------|
| 1       | For Those About To Rock We Salute You | 1        | 1          | AC/DC             |
| 1       | For Those About To Rock We Salute You | 2        | 2          | Accept            |
| 1       | For Those About To Rock We Salute You | 3        | 3          | Aerosmith         |
| 1       | For Those About To Rock We Salute You | 4        | 4          | Alanis Morissette |
| 1       | For Those About To Rock We Salute You | 5        | 5          | Alice In Chains   |

[ ] 1 #INNER Join  
2 %%sql  
3 select \* from album JOIN Artist ON album.artistId=Artist.artistId limit 5;

\* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook\_Sqlite.sqlite  
Done.

| AlbumId | Title                                 | ArtistId | ArtistId_1 | Name      |
|---------|---------------------------------------|----------|------------|-----------|
| 1       | For Those About To Rock We Salute You | 1        | 1          | AC/DC     |
| 2       | Balls to the Wall                     | 2        | 2          | Accept    |
| 3       | Restless and Wild                     | 2        | 2          | Accept    |
| 4       | Let There Be Rock                     | 1        | 1          | AC/DC     |
| 5       | Big Ones                              | 3        | 3          | Aerosmith |

[ ] 1 #SELECT ... FROM alumno JOIN curso USING (aid,...)  
2 %%sql

0s completed at 9:43 AM

# Practicar

[https://github.com/adigenova/uohdb/blob/main/code/SQL\\_IV\\_chinook\\_db.ipynb](https://github.com/adigenova/uohdb/blob/main/code/SQL_IV_chinook_db.ipynb)

# Consultas?

Consultas o comentarios?

Muchas gracias